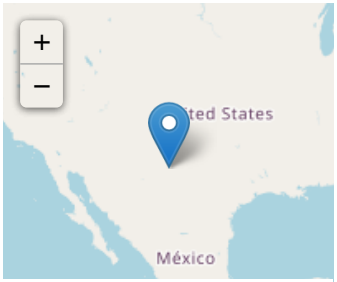
AquaPedia Case Study Database

The Pecos River Compact and Texas - New Mexico Dispute

Case Description



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Geolocation:	32° 0' 0.00" N, 103° 58' 48.00" W
Total Population:	~0.39 million
Total Area:	km^2
Climate Descriptors:	Semi-arid/steppe (Köppen B-type)
Predominent Land Use Descriptors:	agricultural- cropland and pasture, conservation lands, rangeland
Important Uses of Water:	Agriculture or Irrigation,Domestic/Urban Supply,Hydropower Generation
Water Features:	Pecos River;
Riparians:	New Mexico (U.S.); Texas (U.S.);
Agreements:	Pecos River Compact (1948)

Summary

The Pecos River originates in New Mexico and travels through Texas, but highly variant flows have caused a century and more of conflict between the two states. After decades of scrapped plans and congressional plotting, a 1948 compact between the states was created with the goal of finally settling the issue of Pecos water rights. Unfortunately, the compact did not succeed in eliminating controversy. Debates raged over the accuracy of scientific information, the appropriation doctrine and protected uses, and the ability to rework an existing agreement, leading to a protracted Supreme Court lawsuit. Ultimately, New Mexico has had to aggressively curtail its surface and ground water usage in order to comply with the court order, including buying up and retiring irrigated farmland. This case will cover the history of the Pecos River dispute, with the early sections focusing on the bad blood and poor negotiation prior to the 1948 compact, and later sections covering how that history caused the compact to fail through a combination of vagueness and an undeserved trust in a new scientific model.

Natural, Historic, Economic, Regional, and Political Framework

Timeline

- 1821 – first settling, at headwaters

- 1862 – US Army builds Fort Sumner to promote settling in basin
- 1880s – first major basin irrigation projects (independent)
- 1900s – first settlers in the Texas part of Pecos, fail within a decade due to high salt content of water
- 1905 – Bureau of Reclamation begins Carlsbad project, repairing dams, canals, and reservoirs that were failing under private ownership
- 1923 – Texas plans to construct reservoir near New Mexico border in order to support irrigation, formal solution attempted
- 1925 – compact to jointly build Red Bluff dam and reservoir passed by both state legislatures but vetoed by New Mexico governor over insufficient storage for upper river needs (offsetting sedimentation)
- 1926 – New Mexico successfully blocks Texas' attempts to get the federal government to fund a dam at Red Bluff, near the state border
- 1932 – Bureau of Reclamation project ends, stewardship transferred to Carlsbad Irrigation District (CID)
- early 1930s – both New Mexico and Texas petition individually for federal assistance in dam projects (Alamogordo and Red Bluff, respectively) while opposing the other project, and are ordered by Secretary of Interior to reach an agreement or lose both
- 1935 – Alamogordo Agreement is created, indicating both states will agree to each other's dams as long as Texas receives the same proportion of waters from above the Carlsbad projects, with a formal compact to follow
- 1938 – Red Bluff and Alamogordo dams are finished, Texas attempts to officially ratify the Alamogordo Agreement as a compact but New Mexico refuses
- 1939 – as an interim step to the creation of a formal compact, National Resources Planning Board begins Pecos River Joint Investigation (PRJI) of water supply and availability, irrigation, salinity, water use, flooding, erosion, and sedimentation
- 1941 – Royce Tipton becomes the chairman of the PRJI
- 1941 – Texas repeals Alamogordo Agreement to open up possibility of lawsuit in response to flooding and high salinity
- 1942 – Pecos River Joint Investigation publishes results
- 1942 – Texas and New Mexico assign representatives to negotiate a binding agreement on water use (including CID representation)
- 1947 - Royce Tipton joins negotiations a federal representative
- 1948 – Pecos River Compact is signed, ensuring Texas annual water flow in proportion equal to the 1947 flow, not to be reduced by manmade efforts of New Mexico, and establishing the Pecos River Commission to monitor the agreement
- 1951 – a dry year proves impossible for the existing models to calculate the 1947 condition, and Texas disputes the amount of water they receive
- 1957 – the Pecos River Commission begins a Review of Basic Data to update the hydrological models
- 1961 – the Review of Basic Data is released, determining that New Mexico underdelivered a total of 50,000 acre-feet of water since 1948 but manmade reasons were responsible for only 5,000
- 1974 – Texas files suit against New Mexico for failure to deliver water equal to the 1947 levels beginning in 1950 according to the original agreement, in the amount of 1,500,000 acre-feet
- 1988 – Lawsuit resolved, New Mexico must pay 10,000 acre-feet per year in addition to agreed amount, a $14M settlement, and compliance is overseen by a federal River Master
- 1991 – in response to water shortage leading to inability to comply with the revised compact, New Mexico creates the Water Resource Conservation Project, mainly tasked with purchasing and retiring water rights in the basin
- 2001 – Continued failure to meet agreement results in Interstate Stream Commission (ISC) calling a committee of stakeholders to find a long-term solution
- 2003 – ISC and New Mexico agree on plan to purchase land and water rights from CID and the Pecos Valley Artesian Conservancy District, to be pumped in order to meet any shortfall in delivery to Texas

Stakeholder Overview

Stakeholder	Role	Key Issues	Dominant Ideologies
Government of New Mexico	Main player	Quantity of water reserved for New Mexico, Funding for infrastructure	Appropriation doctrine, Efficiency
Government of Texas	Main player	Quantity of water reserved for Texas, Quality of water sent to Texas, Funding for infrastructure	Right of capture, Appropriation doctrine
Carlsbad Irrigation District (CID)	Included party	Quantity of water reserved for New Mexico, Support for agricultural operation in middle basin	Appropriation doctrine
Environmental Groups (PVACD)	Included party	Protection of river ecology and threatened species	Environmentalism, Conservation
Other riparian residents / farmers	Interested party	Quantity of water for personal usage	Right of capture

Ecological and Geographical Background

Summary

The Pecos River originates in New Mexico, traveling roughly south-southeast into Texas before eventually joining the Rio Grande. The river basin includes approximately 44,000 square miles of land, and is split into three general areas: the upper, middle, and lower basins. The upper basin (north of Fort Sumner) includes the headwaters of the river and the southern part of the Rocky Mountains. The middle basin is a valley containing the majority of the population along the Pecos River and the most extensive agricultural activities. The lower basin is the stretch of the river within Texas. The geography of the region (semi-arid plains) makes the flow of the river highly dependent on floodwaters and precipitation from year to year.

Detailed Map

PecosMap.png Source: [Kraai, 1993][1]

River Flow and Key Sources

The flow of the Pecos River varies dramatically along its length and from year to year, to the point where it is difficult to define a meaningful average. This is driven largely by the challenging-to-predict nature of its most important sources: floodwater runoff in the upper and middle basins and two aquifers located in the middle basin. These sources input water to the river at a multitude of locations along its length in amounts that are heavily impacted by the forces of weather and human use, respectively.

The floodwater inflow of the Pecos River is driven by seasonal snowmelt and brief but powerful rainstorms that are collected by numerous small tributaries of the Pecos. These floodwaters can therefore vary considerably between years depending on precipitation; for example, floodwater contributed over 140,000 acre-feet of water in 2010 but only about 60,000 acre-feet in 2012 [2].

Groundwater inflow to the Pecos is provided by springs near the headwaters, an artesian aquifer located near Roswell, and an alluvial aquifer north of Carlsbad. The headwater springs contribute 36,000 to 60,000 acre-feet per year. The amount of water discharged from artesian aquifers is affected by the internal pressure and is thus affected in turn by human intervention through well pumping. Much of the agricultural activity in the middle basin is supported by well-based irrigation, which draws on the aquifer to address shortfall in river-diversion irrigation. When the river runs low due to drought, additional water is drawn from the aquifer, further reducing the inflow to the river. The alluvial aquifer is also filled largely through seepage from the reservoirs created by nearby dams, varying with reservoir level. Well usage and damming practices in the basin have changed considerably over time, but currently the estimated groundwater inflow varies between roughly 20,000 and 40,000 acre-feet per year, though lows of around 15,000 and highs of over 100,000 have been observed as well [3].

Manmade Diversions

Because of the arid climate, significant irrigation projects have been present in the Pecos River basin since the 1880s in order to promote effective agriculture. Multiple dams have been constructed along the river and, early on, were occasionally destroyed by extreme flooding. Currently there are four main dams: the Avalon and Brantley dams created as part of the Carlsbad Reclamation Project in the 1900s, and the Sumner dam (previously known as the Alamogordo dam) near Fort Sumner and the Red Bluff dam located just across the border in Texas. These dams are used primarily to provide water for irrigation projects along the river and to regulate river flow in flood season. Red Bluff also generates hydroelectric power [1].

As previously mentioned, the usage of wellwater to irrigate farmland has had a significant impact on the levels of feeder aquifers to the Pecos, though this was unappreciated until the analysis of the Pecos River Joint Investigation in 1942. Enforcement of the ensuing Pecos River Compact has increased regulation of well usage, but significant water withdrawals from the aquifers are still necessary to make the arid land arable.

Other Natural Considerations

Salt cedars, a notoriously invasive species of tree, thrive on the banks of the Pecos River. These plants are accustomed to highly saline soils and then can aggravate water quantity and quality issues by consuming large amounts of surface water and increasing the salinity of what remains. Efforts to create additional available water for human use in the basin have frequently centered on the reduction or elimination of salt cedar stands, though the impact of these measures has been very unclear.

The Pecos River is also home to a number of endangered species, including the Pecos bluntnose shiner (a small, shallows fish), the interior least tern (a bird that nests on beaches between floods), and a handful of wetlands invertebrates and plants [3][4]. Federal law mandates the protection of the habitats of these creatures, which are endangered partially due to manmade impacts on the flow of the Pecos prior to environmental regulation.

Governmental, Political, and Legal Context

Early Development

The first settlers of the Pecos River basin established themselves in the 1820s, with development of the Pecos River basin beginning in earnest starting in 1862 when the US Army constructed Fort Sumner on the river near where it enters the valley. The express purpose of Fort Sumner was the protection of newly arriving settlers from Native American raids, demonstrating the longstanding federal interest in the growth and prosperity of the region. It is worth noting that, despite the attempted internment of both Navajo and Mescalero peoples near Fort Sumner at the short-lived Bosque Redondo reservation, neither of these tribes had a significant cultural attachment to the Pecos River, as their main territories were located elsewhere and they had returned to those areas by 1870. The 1880s and 1890s featured a burst of privately-funded damming and irrigation efforts in the middle basin by prospecting industrialists, eager to squeeze profits out of the land. Unfortunately, most of these infrastructure companies went out of business due to a combination of the unpredictability of the seasonal Pecos flow leading to considerable farming challenges and the damage and destruction of many dams by fierce floods. The 1900s featured the first attempted settlements of the lower basin, but these also proved fruitless without the support of functional irrigation infrastructure due to the unreliable water flow and considerably alkali water quality in Texas after traveling through the salty soil and salt cedar stands near the border [4].

Because of these challenges, the federal government again invested heavily in the middle basin starting in 1905, when the Bureau of Reclamation began the Carlsbad Project, buying up the failed dams and irrigation canals of the previous decades and repairing them to better support the local farmers rather than the remaining profit-driven corporations in the area. The current Avalon and Brantley dams are direct descendants of this project. The Carlsbad Project officially ended in 1932, with stewardship of the improved irrigation infrastructure passed on to the Carlsbad Irrigation District (CID): a local collective organization charged with maintaining the physical plant for the benefit of the community (though in truth, the Bureau of Reclamation continued to provide considerable support through 1950) [5].

Policy Differences between New Mexico and Texas

Though New Mexico and Texas are not as culturally distinct as the participants in many cross-boundary water disputes between nations or ethnic groups, differences in their state policies have contributed to the underlying tension surrounding the usage of the Pecos River. Firstly, both states share the overarching law of the appropriation doctrine: a means of allocating water rights common to the western United States that gives priority to users based on the order in which they began using it [6][7][8]. Details in implementation can vary from state to state though, which may result in friction when assigning rights between states. The timing of when particular uses officially began becomes a key debate point between Texas and New Mexico in the 1980s.

New Mexico has also pursued efficient usage of its water resources for much of its history, as evidenced by the considerable infrastructure investments. The state was also on the cutting edge of groundwater protection, implementing regulations on the use of groundwater in the 1920s citing the aquifers as “public waters”. This can be viewed in direct contrast to the Texas policy on groundwater that, other than a vague prohibition against “waste”, was fully encompassed by the overlying land rights. This difference has been theorized to contribute to the hostility in Pecos River negotiations, particularly on the part of New Mexico which could consider all of Texas’ policy wasteful [9].

Setting the Stage - The Alamogordo Agreement

Despite the failures of previous settlements along the Pecos River, Texas was still committed to utilizing that land and the Pecos waters. To do so, the state determined that it would need a dam near the New Mexico border in order to improve storage, along with some sort of agreement with New Mexico ensuring a supply of water. New Mexico, always interested in a new opportunity to improve water usage, was also interested in a new dam and reservoir as a means of offsetting sedimentation in other dams of the ongoing Carlsbad Project. A representative for each of the two states and a representative of the Bureau of Reclamation began meetings in 1923 to outline a joint reservoir project at Red Bluff. Information about these negotiations is difficult to find, possibly because (compared to the rest of the history between the states) an agreement was reached relatively quickly. 1925 saw the formal presentation of a compact enacting the Red Bluff project to both state legislatures, where it was passed. However, the governor of New Mexico vetoed the bill under pressure from local irrigators because there was not any explicit language promising additional storage for New Mexico in response to sedimentation [9]. This was the first move in a long line of antagonistic behavior between the states.

Soon after the surprise failure of the 1925 compact, Texas decided to take its claim to the federal government, successfully pressuring Congress into charging the Department of the Interior with the Red Bluff dam project. However, New Mexico interfered with this plan by restricting the project’s financing to a nearly empty and not nearly sufficient fund [9]. Both states would continue to pester the federal government for years about separately building their own pet projects while resisting the other’s proposal. Texas refused New Mexico additional storage on the grounds that they had a more dire need, with no dams of their own. New Mexico opposed Texas’ offerings using the principle of the appropriation doctrine, citing its older settlements. Eventually, the Secretary of the Interior demanded that the states reach an accord or neither project would be funded [10].

To placate the federal government, New Mexico and Texas quickly leapt to create and sign the Alamogordo Agreement in 1935 (through the CID and the Red Bluff Water Power Control District), agreeing to limit the amount of irrigated land along the Pecos in New Mexico and to continue to allow Texas to receive all surplus flow. Additionally, the Agreement explicitly called for the adoption of an official inter-state policy in the near future. The Alamogordo Agreement succeeded in convincing the Secretary of the Interior to allow the construction of both New Mexico’s Alamogordo dam and Texas’ Red Bluff dams [5]. However, this speedy maneuvering had two key problems. First, the Alamogordo Agreement was not really constructed with mutual benefit in mind (other than the construction of the desired dams) and was instead left intentionally vague so that both states could continue to impose their own beliefs on the usage of the Pecos water. Second, the Alamogordo Agreement was signed by state representatives but was not official state law, as the state legislatures were not consulted in its creation, hence the provision to create an official binding agreement later. Texas’ legislature ratified the Alamogordo Agreement, only to later repeal it when New Mexico’s legislature did not. By the year 1938, both dams had been constructed and the incentive to create a formal compact had been removed; yet considerable friction still existed between the states [9].

Negotiating the River - The Pecos River Compact

In 1942, both states again attempted to resolve the allocation of the Pecos River once and for all, sending representative parties to negotiate the terms. The CID was also allowed representation at these meetings, as the largest collective of water usage and irrigation along the river and also the most senior rights holder according to the appropriation doctrine. The negotiations were set to utilize the hydrological data collected by the Pecos River Joint Investigation (PRJI): a study conducted by the National Resources Planning Board from 1939 to 1942 to learn more about the highly complex water dynamics in the basin [1]. The advanced data did not succeed in making negotiations easier. After year of hostile deadlock, the federal government sent the chairman of the PRJI, a prominent hydrological engineer named Royce Tipton, to be a national representative to the meetings in 1947. Tipton would go on to have a massive impact on the shaping of the resulting compact and is considered largely responsible for an agreement being reached at all [10].

Tipton had coordinated the PRJI to focus on the inflows and outflows of the river, constructing an elaborate hydrological model of the basin using the most cutting edge modeling techniques of the time. Correspondingly, the agreement he shaped focused on balancing the available water between the states on a yearly basis as a function of inflows and the infrastructure developments in use. Tipton seized on the idea of destroying the salt cedars to increase the amount of available water, estimating that they consumed up to 56,000 acre-feet per year, and used it to ally Texas and New Mexico behind the same cause [10]. The additional water was used to placate New Mexico and the CID's demands for prior-use protection under the appropriation doctrine as well as Texas' demands for minimum water guarantees, even though the total estimation was likely not enough for both sides. Tipton essentially performed shuttle negotiation between the different groups, privately urging them to follow the models and agree to the developing compact between official meetings by using the worst-case scenario with current usage patterns to provide extra motivation. The unofficial nature of much of these negotiations makes details hard to collect, but Tipton has also been accused of telling each party what they wanted to hear in order to force through an agreement. Eventually, he succeeded in 1948 when the two states signed the Pecos River Compact and it was ratified into law.

The Pecos River Compact had three main outcomes: the establishment of the '1947 condition', an agreement to support legislation for all water-related construction projects in both states, and the creation of a Pecos River Commission to oversee the administration of the compact [11]. The agreement to support construction projects was designed to prevent another deadlock like the Alamogordo versus Red Bluff conflict, instead trusting the compact to maintain the interests of both parties. The key verbiage revolved around the '1947 condition', a guarantee that New Mexico would not conduct any activities that would result in a quantity of water sent to Texas lower than in 1947. The 1947 condition was not an absolute amount, rather it was made with respect to the inflow/outflow models of the PRJI, essentially saying "Texas must receive at least as much water as they would have if this year's floodwaters had occurred given the infrastructure and proportional water uses in 1947 New Mexico." This ensured New Mexico would not be held responsible for extreme weather outside of its control. The creation of additional water, for example through the destruction of salt cedars or improved infrastructure, would thus also benefit New Mexico by allowing them to draw additional water without violating the compact, while Texas was assured a functional quantity of water for their own uses. The Pecos River Commission was, among its other administrative duties, placed in charge of evaluating the 1947 condition in each year. Royce Tipton, as the creator of the models, was an advisor on this task.

Conflict Continues - Texas vs. New Mexico

The Pecos River Compact didn't last without controversy for long. The cutting-edge hydrological models used to describe the 1947 condition were in fact not yet sophisticated enough to accurately describe the basin. A particularly dry year in 1951 proceeded to break the models, with not enough inflow to even calculate the appropriate share for Texas. The most difficult challenge was not determining what Texas was owed given the yearly floodwaters, but rather accurately estimating the floodwaters at all. The Pecos River Commission conducted a Review of Basic Data between 1957 and 1961 to update the models and improve the distribution of water between the states. They concluded that the models had been partially based on unusually wet years in the early 1940s, and then when adjusted to more appropriate levels, New Mexico had delivered a water shortfall of 50,000 acre-feet to Texas and was responsible for only 5,000 acre-feet due to man-made uses [10][1].

Texas was not pleased with this change of affairs, which resulted in significantly less water for their uses. The 1960s were dry, with very little Pecos water making it into Texas despite Tipton's federally funded anti-salt-cedar activities, causing havoc for the farmers of western Texas [10]. In 1974, Texas filed a lawsuit against New Mexico, accusing them of

underdelivering water since 1950. Using the models of the original agreement, this came out to over a million acre-feet of shortage. The resulting lawsuit hinged on the interpretation of a few critical lines of the compact [9][12]:

- What data should be used to calculate the 1947 condition? Texas claimed the original models defined the agreement, while New Mexico claimed that the Pecos River Commission had the authority to update those models to be as accurate as possible.
- Is the 1947 condition defined at the beginning or the end of 1947? Texas argued that it was the beginning, but New Mexico believed that all of its activities by the end of 1947 (including a considerable number of new groundwater wells) should be protected as prior uses.
- How is salvaged water, for example from the elimination of salt cedars, to be accounted? New Mexico thought that all savings from salvage could be exchanged for more groundwater pumping, while Texas argued that the groundwater pumping was limited unilaterally by the compact and both states were entitled to use the salvaged water separately.
- Is New Mexico allowed to pump as much (absolute) groundwater as they were in 1947? Or are they allowed only to pump the amount of groundwater resulting in the same diminishment of the river's flow?

The 'masters' appointed by the Supreme Court to evaluate the facts of the case came to conclusions split between the interests of the two states. The masters determined that the 1947 condition referred to the beginning of 1947, but that it was in fact subject to continual model improvement and that salvaged water could be traded off for additional pumping (no opinion was offered on the fourth point). After fourteen years, a verdict was rendered in 1988, creating a position of federal River Master to the Pecos, charged with overseeing the future delivery of water to Texas in accordance with the amended compact. New Mexico was also made to pay a $14M settlement to Texas and to deliver a minimum extra 10,000 acre-feet of water per year until the total shortfall was met [13]. The masters also suggested that the Pecos River Compact be amended to include an official tiebreaking / grievance-handling procedure, but the Supreme Court ruled that it did not have the authority to make such a change and left it to the states themselves [6].

The Aftermath

New Mexico has not had an easy time meeting the requirements of the amended Pecos River Compact. As early as 1991, New Mexico was forced to create a Water Resource Conservation Project, which outlined a plan to buy up irrigated land and retire it in order to reduce water usage in the middle basin. Again in 2001, a water delivery shortage was anticipated, prompting the New Mexico Interstate Stream Commission (ISC) to organize a meeting of stakeholders to devise a new plan of meeting their obligations [3]. The meeting was quite impressive given the longstanding tensions between many of the attendees, including the State Engineer, ISC, Bureau of Reclamation, CID, and the Pecos Valley Artesian Conservancy District (PVACD). Ultimately, in 2003, another elaborate settlement with a plan to buy up land (and the associated water rights) from the CID and PVACD, with the CID serving as a buffer and the PVACD land only to be pumped to the river in the event of drastic shortage. In return, the CID was allocated ownership of excess water stored in the reservoirs in years which New Mexico could easily meet Texas' needs [14].

Social, Economic, and Cultural Context

Being a dispute largely between two states of the USA, the debate over usage of the Pecos River has not been significantly driven by socioeconomic or cultural factors. The population along the Pecos in both New Mexico and Texas are of similar descent and are both heavily involved with the agriculture industry.

Issues and Stakeholders

Water Quantity (NM)

NSPD: Water Quantity, Ecosystems
Stakeholder Types: Federated state/territorial/provincial government, Local Government, Environmental interest, Community or organized citizens

New Mexico needs considerable quantities of water in order to support the extensive irrigation that allows agriculture in the middle basin. This water has come from the surface water of the Pecos (drawn from dam reservoirs) and groundwater withdrawals from the aquifers that feed the Pecos. The state government also has received considerable pressure on this issue from farmer collectives such as the Carlsbad Irrigation District and environmental groups such as the Pecos Valley Artesian Conservancy District, which strongly oppose any reduction in water quantity allocated to New Mexico. These stakeholders demand support for local agriculture and protection of endangered species and natural ecology, which require quantities of water. Existing uses of the water such as these are very high, and any order to increase downstream flow requires that New Mexico strategically retire current (and potentially longstanding) uses.

Water Quantity and Quality (TX)

NSPD: Water Quantity, Water Quality
Stakeholder Types: Federated state/territorial/provincial government, Community or organized citizens

Texas also requires water from the Pecos River to support agriculture, but lacks the benefit of the groundwater provided by aquifers and is forced to use the Red Bluff reservoir for most of its water needs. Red Bluff also generates hydropower for the area, which is dependent on water quantity. Furthermore, the water quality of the Pecos becomes gradually more saline as the river approaches Texas, which has presented difficulties for Texan farmers in the past. Solutions that would further deteriorate water quality are essentially untenable for Texas.

Infrastructure Funding

NSPD: Assets
Stakeholder Types: Federated state/territorial/provincial government, Sovereign state/national/federal government

Both states have, in the past, petitioned for (and received) considerable funding and assistance from the federal government. The Alamogordo and Red Bluff dams are direct results of the funding, while older dams like the Avalon and Brantley dams are legacies of the Bureau of Reclamation's presence in the early settling of the area. The Pecos River Compact largely put this issue to bed, as it has not been a prominent source of conflict in recent years, but the early history of the Pecos River is littered with funding challenges for both Texas and New Mexico, who aggressively tried to claim limited federal funds for themselves and block funding for the other. This issue could return if further infrastructure is requested.

Agricultural Support

NSPD: Water Quantity, Values and Norms
Stakeholder Types: Federated state/territorial/provincial government, Local Government, Community or organized citizens

The Carlsbad Irrigation District (and other, less important farming collectives) look to the government for protection, using their position as a key economic driver and livelihood for a significant percentage of the local population as a normative argument for why they should be served. Because of the water-intensive irrigation necessary to grow crops in the American southwest, these stakeholders are constantly looking to store surface water behind dams and draw groundwater from aquifers. Reductions in water allocations often require that farmland be retired. The CID was invited to participate in the Pecos River Compact negotiations with voting representation due to their significant influence; many other districts and local farmers in both Texas and New Mexico were limited to observing and participating in occasional open forums for water issues intended to feed information to state representatives.

Environmental Protection

NSPD: Ecosystems, Values and Norms
Stakeholder Types: Federated state/territorial/provincial government, Environmental interest

The Pecos River has been the habitat for a number of endangered or threatened species of plants and animals, in some cases endangered due to environmental changes from early 1900s river diversions. Early policy in New Mexico treated water as a limited resource to be managed efficiently but exploited for human use, while policy in Texas was essentially nonexistent, leading to considerable ecological damage. As environmentalism became legislated, environmental groups such as the Pecos Valley Artesian Conservancy District became more powerful stakeholders. In particular, New Mexico has had to work with the PVACD to accommodate the 1988 Supreme Court ruling without further damaging the environment.

Analysis, Synthesis, and Insight

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Key Questions

Transboundary Water Issues: *What kinds of water treaties or agreements between countries can provide sufficient structure and stability to ensure enforceability but also be flexible and adaptable given future uncertainties?*

The Pecos River Compact was designed using a hydrological model based on yearly inflows and outflows rather than fixed quantities, allowing it to continue to apply as environmental conditions changed. Also (though it was a matter of significant contention for Texas) the Compact also allowed room for that model to be improved/updated in later years to make the accounting as accurate as possible.

Transboundary Water Issues: *What mechanisms beyond simple allocation can be incorporated into transboundary water agreements to add value and facilitate resolution?*

Removal of unwanted invasive species (in this case, salt cedars) can be used to liberate additional water for the use of all parties in the negotiation.

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